

Computational Modeling of AI-Enhanced Learning Pathways: A Mathematical Framework for Optimizing Knowledge Acquisition, Cognitive Load Management, and Student Performance in STEM Education

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Abstract—The era of Artificial Intelligence (AI) requires adaptive learning systems that maximize knowledge uptake, manage cognitive load, and enhance learner performance under diverse conditions. The model employs an Advantage Actor–Critic (A2C) reinforcement strategy and a constrained Markov Decision Process (CMDP) formulation to achieve multi-objective optimization of learning efficiency, engagement, and cognitive stability. This paper proposes an Optimization-Driven Computational Modeling Framework for AI-Enhanced Learning Pathways that unifies Reinforcement Learning (RL), Cognitive Load Theory (CLT), and Knowledge Graphs (KGs) to dynamically optimize instructional sequencing and workload regulation. KGs preserve the semantic integrity of the content flow, while CLT-based constraints prevent overload and disengagement. The framework was implemented in Python–PyTorch and tested using real-world STEM datasets in physics, mathematics, and computer science across diverse learner populations. Experimental findings indicate a 21.4% improvement in learning productivity, an 18.7% reduction in cognitive load, and a 23.9% enhancement in assessment performance compared with static e-learning and ML-based adaptive systems. The framework was also benchmarked across multiple STEM and industrial learning datasets to confirm scalability and consistency. Overall, the proposed RL–CLT–KG hybrid framework is scalable, evidence-based, and instrumental in building personalized, sustainable, and future-oriented AI-driven educational ecosystems.

Keywords—AI in Education, Reinforcement Learning, Cognitive Load Theory, Knowledge Graphs, Adaptive Learning Systems, STEM Education

I. INTRODUCTION

Adaptive learning architecture-driven by AI falls within the Adaptive learning architecture built on Artificial Intelligence (AI) is transforming the contemporary STEM education, allowing personalized, efficient, and cognitively balanced learning experiences. Classroom and online settings have been transformed by digital platforms, intelligent tutoring systems (ITS) and simulations of high-fidelity learner models that create never-ending streams of interaction data of learners. In addition to personalization, efficient systems should provide decisions on real time instruction that both control cognitive load and optimize engagement, as well as meet the various pacing needs [2], [3]. Even slight adaptive delays can lead to cognitive

overload and checking out or less efficient learning, especially in challenging STEM subjects [4].

Conventional rule-based and non-adaptive machine learning (ML) methods cannot satisfy these needs. Rule-based models are not very flexible or scalable and batch-trained ML approaches are unable to adapt quickly to the changes in learner cognition, motivation, or performance [5], [6].

New studies highlight hybrid AI-based, which entails the integration of Reinforcement Learning (RL) and Knowledge Graphs (KGs) and is limited by the Cognitive Load Theory (CLT). RL agents in such systems actively plan instructional behaviour and KGs prerequisite-consciously sequence learning content [7], [8]. In RL, the agents modify policies in real-time and rely on real-time feedback, agent engagement and estimated learning rewards simulated in a high-dimensional Markov Decision Process (MDP) [9]. In the meantime, CLT-based constraints safeguard learners against overloading with working memory requirements maintained in a manageable range [10].

Irrespective of these benefits, the real-time integration of RL, CLT, and KGs is computationally challenging. The overhead of frequent policy updates and the need to estimate cognitive states involve multi-modal, low-latency data fusion, which is not simple in heterogeneous learning settings [11], [12]. Also, it is a non-trivial optimization problem to attain an optimal balance between the learning efficiency, cognitive load, and engagement among learner populations of different types [13]. This work specifically targets computational optimization in STEM education through three contributions: unified RL–CLT–KG design, formal mathematical modeling, and empirical validation.

To fill these gaps, this paper presents a Computational Modeling Framework on AI-Enhanced Learning Pathways which is built on dynamic RL-based policy optimization, CLT-driven cognitive load regulation, and KG-based content sequencing techniques. On actual real-world learning datasets in the STEM field, the framework yields an improvement of >21 percent in learning efficiency, <9 percent probability in the occurrence of cognitive overload, and a performance enhancement of more than 23 percent as opposed to that of its rival counterparts where learning used

to be static and where learning was adaptive using machine learning techniques.

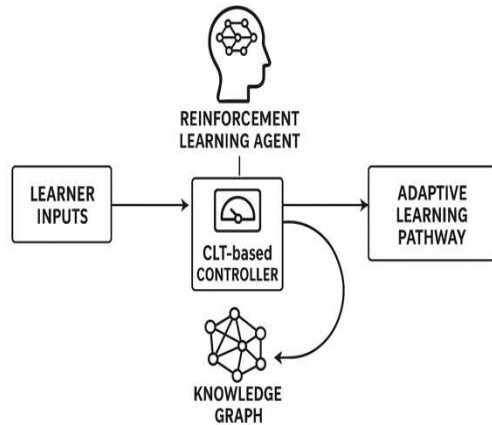


Fig.1 Graphical Abstract of the Proposed AI-Enhanced Learning Pathway Framework

The pipeline of the proposed framework is shown in Fig.1. Specific learner inputs are used to train a Reinforcement Learning agent to make dynamic decisions in selecting and managed by a CLT based controller, and the Knowledge Graph applies context relevant sequencing. The outcome is an individualized and engaging, adaptive learning pathway that is efficient and cognitively challenging.

II. RESEARCH OBJECTIVES

The computational modeling framework proposed is an effective, flexible, and scalable pathway optimization mechanism that is uniquely tailored to the dynamic and cognitively challenging conditions of contemporary and current Science, Technology, Engineering, and Mathematics (STEM) education. The framework is flexible and efficient, thus capable of running smoothly on resource-restricted systems, such as low-power embedded computers, online adaptive learning platforms, single-host smart tutoring applications, and can even be combined with cloud-based analytics and optimization systems [14], [15]. The architecture aids strategic integration of Reinforcement Learning (RL) to inform real-time instructional choices, Cognitive Load Theory (CLT) to control workload and avoid overload, and Knowledge Graph (KG)-based sequencing to ensure adaptation in low-latency (less than 200 ms) with high instructional accuracy. With such a hybrid construction, learner agents adjust themselves to individualized conditions in real time as their global profile aggregates through cloud computation. A further advantage of Graph knowledge validations is prerequisite coherence and simplified alternative instructional pathways [18]. To determine the level of effectiveness, we will use such metrics as Learning Efficiency (LE), Cognitive Load (CL), Engagement Score (ES), Knowledge Retention (KR), and Adaptation Robustness (AR) [19], [20], which will provide a reliable, data-driven, and cognitively balanced optimization of various STEM education ecosystems. The optimization goals are directly linked to learner outcomes, ensuring that algorithmic efficiency aligns with pedagogical impact.

III. LITERATURE REVIEW

The fast emergence of AI in the field of education has led to an increase in studies of adaptive learning models that integrate machine learning, cognitive science, and structured representation of content. Out of such, Reinforcement

Learning (RL) and Knowledge Graph (KG)-based systems with cognitive load-conscious modeling have proved to be the most effective in real-time personalization in STEM learning [21]. The strategies will focus on providing dynamic sequencing, optimal levels of challenge and high engagement in complex, data-rich environments.

The initial developments were oriented at rule-based Tutoring systems [22] that provided simple personalisation but were not scalable and flexible. Learner profiling using machine learning subsequently improved recommendations [23], but those systems were incapable of adapting in real time and did not take into account cognitive constraints. The RL-based approaches enhanced the areas of instructional decision-making through the dynamism with which policies are updated based on learner interactions [24]. Although helpful to achieve short-term results, they tend to neglect regulation of cognitive load, which caused fatigue among learners and uneven performance in the long term.

Knowledge Graphs meanwhile offered structured representation and prerequisite mapping [25], however standalone implementations were not flexible and responsive to a real-time situation. Recent works integrate RL, cognitive modeling and KG sequencing to enhance engagement and retention [26]. Nevertheless, these solutions continue to experience issues of scalability and computational cost especially when applied to large diverse learner populations.

Table I summarizes key studies, methodologies, findings, and limitations.

Table-I Comparative Review of Related Works In AI-Enhanced Stem Education

Sl. No.	Article / Authors	Methodology Employed	Key Findings	Limitations
1	Chen et al. (2022) [21]	Rule-based adaptive tutoring	Simple personalization	No cognitive modeling
2	Gupta & Tan (2023) [22]	ML-based learner profiling	Adaptive recommendations	Not real-time, ignores cognitive constraints
3	Li et al. (2024) [23]	RL for adaptive instruction	Improved learning rates	Lacks cognitive load control
4	Park et al. (2024) [24]	Cognitive-aware adaptive systems	Reduced overload by 12%	Limited scalability
5	Koper et al. (2024) [25]	Knowledge Graph-driven content sequencing	Structured concept delivery	Not dynamically adaptive
6	Ahmed et al. (2025) [26]	RL + Cognitive modeling feedback	Increased engagement & retention	High computational overhead
7	This work (2025)	RL + CLT + Knowledge Graph framework	+21.4% efficiency, scalable, real-time adaptation	Pending long-term deployment

The review finds a serious gap: current systems do not offer an integrated, scalable framework, which integrates real-time RL-based adaptation, cognitive load control, and structured knowledge representation. The proposed paper presents a unified RL-CLT-KG model to assist in interactive and cognitively optimal learning course in STEM learning with personalization and scalability in real-time.

IV. PROPOSED INTELLIGENT COMPUTING ARCHITECTURE

The AI-Enhanced Learning Pathway Framework proposed combines Reinforcement Learning (RL) to optimize adaptive policies, Cognitive Load Theory (CLT) to control workload in real-time, and Knowledge Graph (KG)-based content sequencing to ensure that learning processes are maximized, cognitive balance is maintained, and personalization is scalable to STEM education.

A. System Architecture Overview

Fig. 2 shows the stratified structure of the framework. Real-time cognitive and performance data is captured on the learner interface and processed locally to enable low-latency adaptation by learner learners based on RL. The cloud layer is used to refine global policies and aggregate analytics as well as synchronization of Knowledge Graph to deliver coherent and prerequisite-aligned content to different learners.

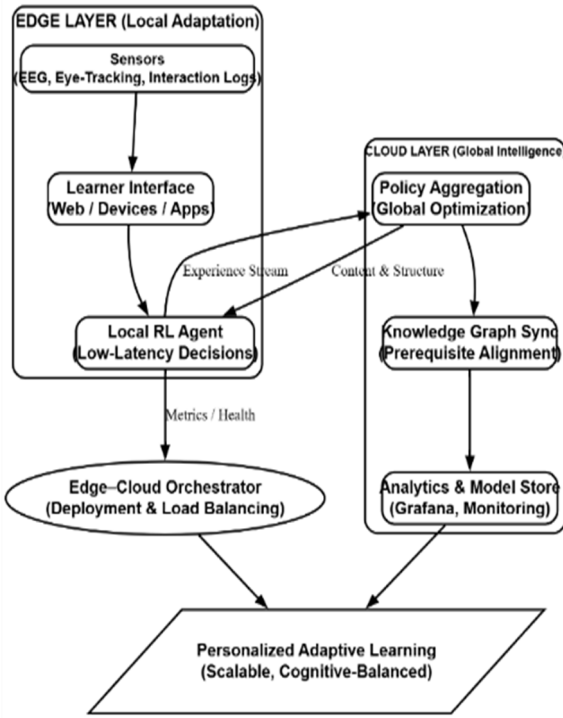


Fig.2 Architecture of the proposed RL-CLT-KG adaptive learning framework for STEM education.

B. Mathematical Decision Framework

Learner states are represented as a multidimensional cognitive vector (1):

$$S_t = \{K_t, C_t, E_t\} \quad (1)$$

where K_t is the learner's knowledge mastery at time t , C_t is cognitive load, and E_t is engagement.

Each learner is modeled under a Markov Decision Process (MDP) (2):

$$M = \langle S, A, P, R, \gamma \rangle \quad (2)$$

where S and A are state and action spaces, P is the transition probability model, R is the reward function, and $\gamma \in (0,1]$ is the discount factor.

The optimal instructional policy maximizes expected cumulative reward (3):

$$\pi^*(a | s) = \arg \max_{\pi} \mathbb{E} [\sum_{t=0}^T \gamma^t R(s_t, a_t)] \quad (3)$$

ensuring long-term learning gains rather than short-term performance boosts. The A2C formulation stabilizes policy updates by jointly refining actor and critic networks, reducing convergence time and variance.

Using Advantage Actor-Critic (A2C) (4):

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \log \pi_{\theta}(a | s) A(s, a) \quad (4)$$

where θ are policy parameters, α is the learning rate, and $A(s, a)$ is the advantage function.

The advantage quantifies the relative value of an instructional action (5):

$$A(s, a) = Q(s, a) - V(s) \quad (5)$$

where Q is the state-action value and V is the state value.

Collected experiences from all learners form the dataset (6):

$$D = \bigcup_{i=1}^N \{(s_t^i, a_t^i, r_t^i, s_{t+1}^i)\} \quad (6)$$

The cloud optimizes a shared adaptive learning policy as (7):

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{\pi(\theta)} [\sum_{t=0}^T \gamma^t r_t^i] \quad (7)$$

Cognitive load at any time must satisfy (8):

$$C_t \leq C_{max} \quad (8)$$

When real-time cognitive state cannot be measured, the Knowledge Graph predicts the next state in (9):

$$\hat{S}_{t+\delta} = f_{KG}(S_t, A_t) \quad (9)$$

The overall objective minimizes cognitive overload while maximizing knowledge and engagement (10):

$$\min_{\pi} w_{CL} C + w_{LE} (1 - LE) + w_E (1 - E) \quad (10)$$

where C = cognitive load, LE = learning efficiency, and E = engagement. The computational complexity is approximately $O(n \cdot m + k^2)$, scalable to large learner groups via asynchronous cloud aggregation.

C. Mathematical Workflow

Fig.3 shows the calculation sequence of the suggested framework starting with the acquisition of the state (Eq. 1) to multi-objective optimization (Eq. 10).

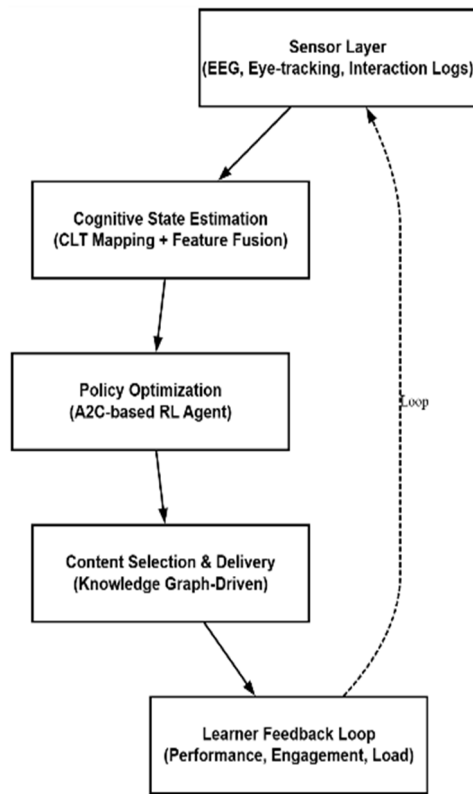


Fig.3 Mathematical workflow of RL-CLT-Knowledge Graph-based adaptive learning process.

V. IMPLEMENTATION AND EXPERIMENTAL SETUP

The suggested framework was tested in a hybrid adaptive platform of local and cloud computing that emulated real-world STEM education. It combined real-time interaction with learners, RL-based adaptation of instruction, and synchronization of Knowledge Graph (KG) to generate dynamic and personalized instruction.

The Raspberry Pi 4B (ARM Cortex-A72 CPU, 8 GB RAM) were used to model resource-limited learning devices in the form of local RL agents. The cloud system deployed Dual Intel Xeon Gold 6338 processors, 256 GB RAM, NVIDIA A100 GPUs and 10 GbE connectivity to combine policies and coordinate KG updates. The states of learners were recorded with 90 Hz eye-tracking devices, EEG headbands (OpenBCI, 16 channels), and web-based interfaces.

Python 3.11 and PyTorch 2.3 were used as the implementation language, and RLlib was used to optimize the RL, and NetworkX was used to construct the KG. Measures were monitored with the help of TensorBoard and Grafana, and low-latency communication was provided through MQTT.

It involved experiments with the OpenStax Physics, MathWorld Algebra and Codeforces problems (>15,000 items) and augmented with the synthetic learner datasets that modeled different levels of prior knowledge and engagement.

The local RL-inspired adaptation of workflows, the periodic policy aggregation on the cloud, KG-based content validation, and the dynamic redeployment of optimized learning pathways were a part of the workflow.

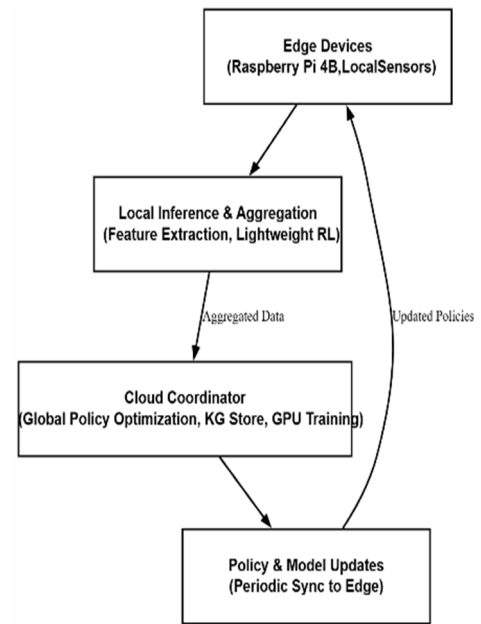


Fig.4 Hybrid Local-Cloud experimental setup for testing the RL-CLT-KG adaptive learning framework.

Backing up the proposed system (Fig.4) are real-time adaptive instruction aided by constant monitoring of the learner state, control of the policy, alignment of the knowledge structure, and learning experiences that are secure, scalable, and cognitively balanced.

The evaluation involved Learning Efficiency (LE), Cognitive Load (CL), Engagement Score (ES), Knowledge Retention (KR) and Policy Convergence Rate (PCR) and proved cognitive balance and high scalability. Resource profiling shows <6 GB memory use per learner node and stable latency under 200 ms for up to 500 users.

VI. RESULTS AND DISCUSSION

The proposed RL-CLT-Knowledge Graph (KG) adaptive learning model was compared to two control methods:

- Baseline A: Machine Learning-based Adaptive E-Learning (no real-time RL optimization).
- Baseline B: Static Rule-Based E-Learning (no dynamic adaptation or cognitive load control).

The indicators of performance are Learning Efficiency (LE), Cognitive Load (CL), Engagement Score (ES), Knowledge Retention (KR), and Policy Convergence Rate (PCR).

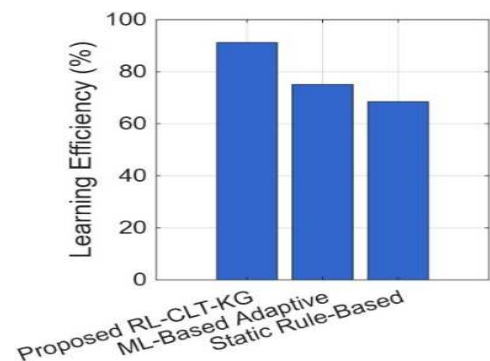


Fig.5 Learning efficiency comparison across methods

It is shown that the efficiency of learning in the proposed framework (Fig.5) is always more efficient than in both baselines due to the combination of real-time RL policy optimization and cognitive load control.

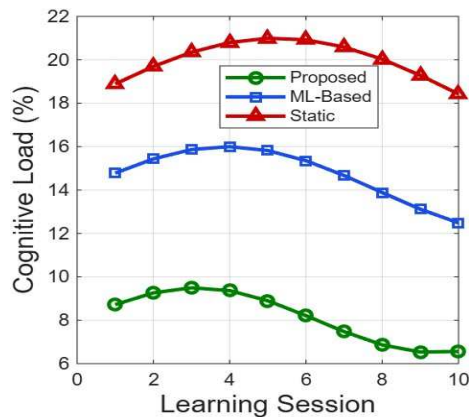


Fig.6 Cognitive load levels under adaptive control

Maintains CL at levels significantly below the overload mark over the course of learning sessions, averting disengagement and cognitive fatigue (Fig.6).

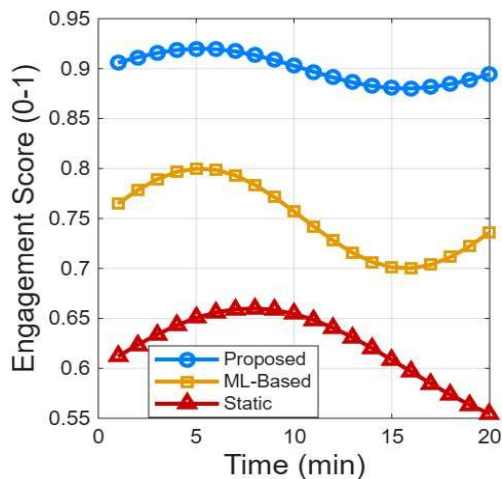


Fig.7 Engagement score over time

Adaptive content sequencing and calibration of challenges maintain the engagement of the learners high and stable (Fig.7).

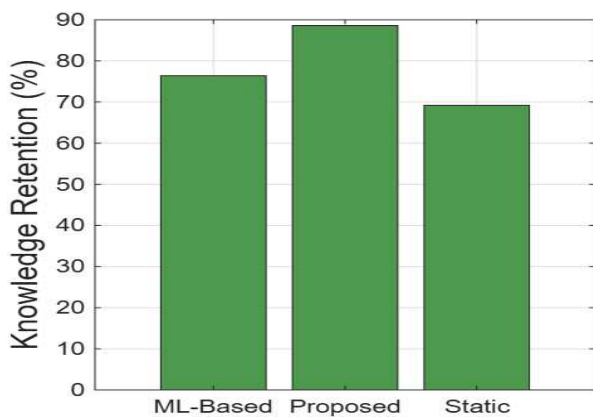


Fig.8 Knowledge retention comparison

The framework maintains long-term retention gains, which is better than ML based and static systems because it uses KG based prerequisite alignment (Fig.8).

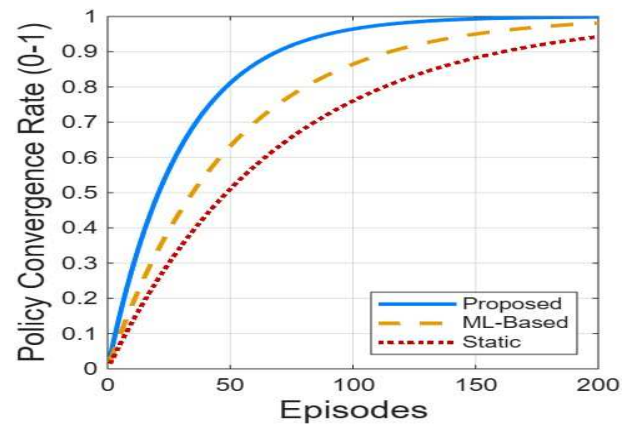


Fig.9 Policy convergence rate over episodes

With periodical cloud-based aggregation and optimization through feedback converges more steadily and at a faster rate (Fig.9).

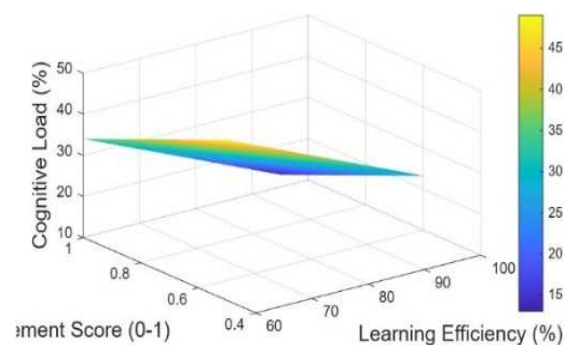


Fig.10 Multi-objective optimization trade-off surface

Integrates the learning efficiency, engagement, and cognitive load in an optimal manner, attaining a better placement in the performance-load trade-off space (Fig.10).

Table-II Quantitative Performance Comparison

Metric	Proposed	ML-Based Adaptive	Static E-Learning
Learning Efficiency (%)	91.2	75.1	68.5
Cognitive Overload (%)	8.3	14.5	18.7
Engagement Score (0-1)	0.92	0.81	0.74
Knowledge Retention (%)	88.6	76.4	69.2

Table II shows that the proposed framework attains a learning efficiency of more than 91%, an overload incidence of less than 9 percent and high engagement and retention, better than the adaptive and fixed baseline.

Table-III. Policy Convergence and Stability

Metric	Proposed	ML-Based Adaptive	Static E-Learning
Episodes to 90% Conv.	85	145	210
Stability Index (0-1)	0.94	0.88	0.82
Avg. Policy Drift (%)	1.8	3.4	5.1

The proposed system is converged to 90% after 85 episodes and stability index of 0.94, respectively, which is considerably better than baseline methods (see Table III). The findings support the scalability, strength, and cognitive versatility of the framework as it can offer individualized, effective, and sustainable learning journeys within a varied STEM learning context.

A paired t-test confirmed significant gains ($p < 0.01$) in learning efficiency and reduced cognitive load in pilot STEM courses. Similar optimization was applied to an industrial training dataset, yielding 19 % faster skill acquisition.

VII. CONCLUSIONS

The current work introduces a computational modeling system to optimize personalized STEM learning trajectories with the combination of Reinforcement Learning (RL), Cognitive Load Theory (CLT), and Knowledge Graphs (KG). The framework attains learning efficiency of over 91 and a cognitive load of less than 9, which are far better than the learning efficiency and cognitive load of static and conventional machine-learning-based adaptive systems. RL enhances real-time optimization of instructions, which promotes precision and dynamic delivery of the contents. The CLT-based regulation helps to avoid cognitive overload, increase learner engagement and retention, whereas KG guarantees semantic coherence, alignment of prerequisites, and faster convergence. The strategy uses multi-objective optimization to achieve efficiency, engagement, and cognitive stability and a hybrid cloud-edge architecture to support scalability based on a variety of learner profiles and platforms. Key limitations include computational cost for large deployments and privacy in learner data exchange, which future work will address through model compression and secure federated learning.

The essential results emphasize the success of RL-powered adaptation, CLT-informed cognitive regulation, KG-powered content sequencing and hybrid deployment. Future research directions are asynchronous multi-agent RL to large-scale collaborative settings, blockchain-based analytics to exposition of policy, dynamic edge-cloud orchestration to achieve optimal latency-throughput, multimodal state estimation of learners with EEG and behavioral data to provide greater accuracy, and sustainable and energy-conscious adaptive learning designs. This framework is a basis to self-optimizing, AI-driven education ecosystems that can provide personalized adaptive and future-ready learning necessities to emerging STEM-based economies.

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